COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
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ACTIVITY NO. 6.1 **Linear and Binary Search**

1. PROCEDURE OUTPUT

1. What is a search tree in data structures?

A search tree is hierarchical data that is similar to an actual tree. But instead of leaves and stems, it has nodes and branches. These nodes contain different information that stems into other parts. For example, each node contains data, and that data stems into three more data types, and that data contains other data which represents a tree in a way. The original node from where everything stems from is called the **root** and the nodes that stem from that original node is called the **child**.

2. What are the different types of search algorithms in data structures? Differentiate each type of search.

Linear search - Allows searching even if data is not sorted. It is also called the sequential search. Multidimensional arrays can be used. Linear search performs equality comparisons. It is less complex but slower.

Binary search - Allows searching only if the data is sorted. It is also called the half-interval search. Only single dimensional arrays are used. Binary search performs ordering comparisons. It is more complex and much faster.

3. What operations / implementations can be performed using binary and linear search operations?

Unsorted Data Retrieval - finding the position of an object in a series of data (linear search)

Multi-Match Collection - finding all the occurrences of a specific item in a list (binary search)

4. What are the advantages in using binary search tree as data structure?

Binary Search Tree allows you to dynamically access the memory or data. It stores data in an orderly manner. You can easily search, insert and delete nodes without having to restructure entire memory blocks. It also scales naturally with the data set. Hence, it is called dynamic.

5. Give an example program using binary search and Linear search.

Online Shopping Filters (Unsorted Data) [Linear Search]: If you search for an item in a shopping app, then check for the description of the products such as the colors, sizes and types. The system will perform a linear search to find items that satisfy your checklists.

Network Routing (IP Lookup) [Binary Search]: When loading a website, internet routers have to look for your IP address in order to send your data packets. It uses a system called “longest prefix match” which relies heavily in binary search trees to route traffic globally in milliseconds.

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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